

AI-IoT Based Smart Fertilizer Spraying System for Plant Disease Severity Management

Abstract

The “AI-IoT Based Smart Fertilizer Spraying System for Plant Disease Severity Management” is designed to bring precision agriculture into everyday farming. The system continuously monitors crop and soil conditions using an integrated network of IoT sensors that capture soil moisture, pH, NPK content, temperature, humidity, and high-resolution leaf images. Collected data are transmitted to a cloud platform, where an AI engine—powered by convolutional neural networks (CNNs)—analyzes plant images to identify diseases and evaluate their severity levels in real time.

Based on the detected disease stage and current soil nutrient profile, a decision model computes the precise fertilizer or pesticide dosage required. An automated spraying mechanism, controlled by microcontrollers and IoT-enabled relays, applies the calculated amount directly to affected areas, minimizing chemical waste and human intervention. Farmers receive live updates, historical trends, and actionable alerts through a mobile or web dashboard, enabling data-driven crop management.

This system reduces fertilizer usage by up to 30%, lowers production costs, and limits environmental impact while improving crop yield and disease management. By combining artificial intelligence with Internet of Things technology, the proposed solution offers a scalable, sustainable approach to smart farming and early plant disease control.